

MATH 130: 9/24 WORKSHEET

Last time we talked about *truth tables*, which express the meaning of sentences in propositional logic by exhaustively listing their behavior for any possible combination of inputs. This device lets us take previous definitions and turn them into precise questions we can check by looking at truth tables. This is convenient because it gives us procedures we can use; checking logical properties becomes a matter of turning a crank.

We start with our most guiding concept. Next class we turn to more concepts.

Sentences $P_1, P_2, \dots, P_n$ *entail* a conclusion C if whenever all the premises are true then C must also be true. An argument is *valid* if its premises entail its conclusion.

These concepts can be turned into questions about truth tables by understanding that when we talk about possible worlds (possible states of affairs for what can be true or false) we are talking about rows in a truth table.

Procedure to check validity of an argument. To check validity of an argument

- P_1
- P_2
- $\therefore C$

do the following:

- (1) Make truth tables for P_1 , P_2 , and C .
- (2) If C is true in every row where both P_1 and P_2 are true, then the argument is valid.
- (3) Else, if there is a row where both P_1 and P_2 are true but C is false, then the argument is invalid.

With a different number of premises you proceed similarly. Look for rows where all premises are true.

With this procedure, we can look at common argument patterns and identify which of them are valid and which are logical fallacies. For example, here are four argument patterns about if-then statements.

Modus ponens

- $A \Rightarrow B$
- A
- $\therefore B$

Affirming the consequent

- $A \Rightarrow B$
- B
- $\therefore A$

Modus tollens

- $A \Rightarrow B$
- $\neg B$
- $\therefore \neg A$

Denying the antecedent

- $A \Rightarrow B$
- $\neg A$
- $\therefore \neg B$

Truth tables can tell us which of these are valid and which are invalid.